Vintage Film Camera Equipment

Dual-Input Priority-Switched Buck-Boost Power Adapter

Custom Datasheet & Design Calculations

Document ID	CPA-DD-001
Revision	0.1 — Draft / Template
Date	October 5, 2026
Author	**[TODO: name]**
Status	Work in progress — sections marked **[TODO:]** are incomplete
Related spec	`Specifications/SpecSheet.pdf`
Supporting docs	`BuckBoostTopology.md`, `ComponentSelection.md`, `SchematicBOM.md`

This document defines the board as a standalone power module (custom datasheet) and records the calculations behind every selected value (design calculations).

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1 Introduction

1.1 Purpose

This document serves two functions:

1. A **custom datasheet** for the assembled adapter — what the board does, its limits, and how to connect it.
2. A **design calculation record** — every value on the schematic is derived from an equation traceable to a manufacturer datasheet.

1.2 Scope

The adapter accepts two independent 11 V to 36 V DC inputs with fixed priority (Input 2 preferred), protects against reverse polarity and overcurrent, and delivers a jumper-selected 12 V or 24 V output rated 6 A to 8 A (burst duty cycle; see subsection 3.3).
[TODO: Confirm scope wording against the final schematic.]

1.3 Related documents

Document	Contents
`Specifications/SpecSheet.pdf`	Customer requirements / acceptance criteria
`BuckBoostTopology.md`	LT8390 stage build guide and audit log
`ComponentSelection.md`	Value-by-value selection rationale
`SchematicBOM.md`	Current bill of materials and placement status
`Datasheets/`	Manufacturer datasheets (local copies)

Table 1: Reference documents.

1.4 Acronyms

Term	Meaning
UVLO	Under-voltage lockout
CC	Constant-current (limit)
HS / LS	High-side / low-side MOSFET switch
PGOOD	Power-good output of the LT8390 (active low)
NTC	Negative temperature coefficient thermistor

Table 2: Acronyms and definitions.

2 System Overview

2.1 Block diagram

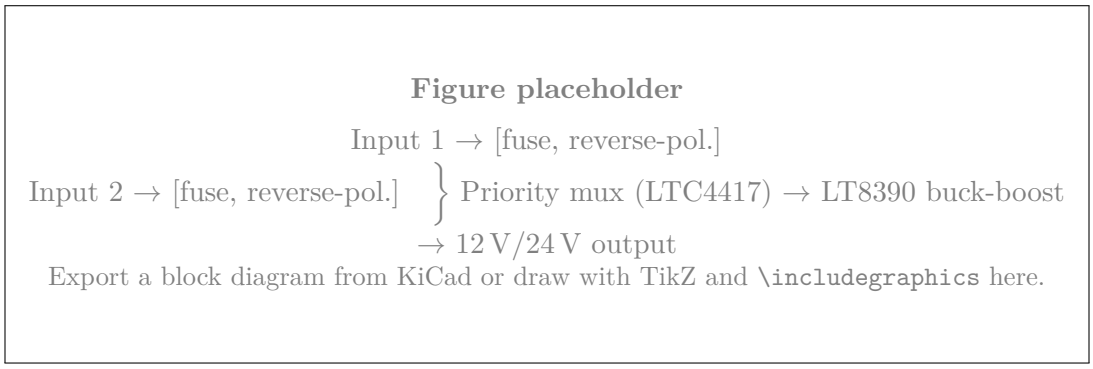


Figure 1: System block diagram. **[TODO: replace placeholder.]**

2.2 Operating modes

Mode	Behaviour
Input priority	Input 2 powers the board whenever it is valid; automatic fallback to Input 1.
Buck region	$V_{IN} > V_{OUT}$ — buck leg switches.
Boost region	$V_{IN} < V_{OUT}$ — boost leg switches.
Overlap band	$V_{IN} \approx V_{OUT}$ — both legs switch (worst thermal corner).
Output select	Jumper fitted = 12 V; jumper removed = 24 V.
Over-temp fault	Latch pulls EN/UVLO low until input power is cycled.

Table 3: Operating modes.

3 Custom Datasheet

3.1 Description

Dual-input, priority-selected, non-isolated synchronous buck-boost DC/DC adapter for vintage film camera equipment. Intended for low-duty-cycle loads (short bursts) inside a small enclosure — the enclosure floor acts as the heat sink through a bottom-side thermal pad.

3.2 Absolute maximum ratings

Parameter	Conditions / note	Max
Input voltage, either input	Continuous; reverse-polarity block extends to −42 V	36 V
Output current	Burst rating	8 A
Output current	Continuous duty (thermal-limited)	**[TODO:** 2 A**?]**
Operating ambient temperature	**[TODO: state enclosure assumption]**	**[TODO:]**
Storage temperature		**[TODO:]**

Table 4: Absolute maximum ratings. Stresses beyond these may damage the board.

3.3 Recommended operating conditions

Parameter	Conditions / note	Value
Input 1 voltage	Valid operating window (UVLO ≈ 11 V)	11 V to 36 V
Input 2 voltage	Valid operating window (UVLO ≈ 11 V)	11 V to 36 V
Output voltage	Jumper selection	12 V or 24 V
Output current	Real duty cycle (per spec)	2 A to 4 A typical
Output current	Short-burst rating	6 A to 8 A
Ambient	Inside camera enclosure	**[TODO:]**

Table 5: Recommended operating conditions.

3.4 Electrical characteristics

[TODO: fill min/typ/max columns once the first board is measured; values below are design targets, not tested limits.]

Parameter	Conditions / note	Target
Inputs		
Input priority	Fixed, not voltage-based	Input 2 > Input 1
UVLO per channel	Set by priority controller; below mux floor	11 V ± **[TODO:]**
Reverse polarity	Both inputs independently	−42 V capable
Fuse per input line	Sized from worst-case input current	**[TODO: rating]**
Input current (worst case)	$P_{OUT}/(\eta \cdot V_{IN,min})$	**[TODO: value in A]**
Output		
Output voltage accuracy	LT8390 FB accuracy	±1.5 %
Output current limit	$R_{11} = 12\,\mathrm{m}\Omega$ **(provisional)**	8.33 A
Output ripple	Measured at rated burst load	**[TODO:** $\mathrm{mV_{pp}}$**]**
Protection & indicators		
Over-temperature trip	NTC/TL431 latch **(provisional)**	≈85 °C ± 8 °C
Fault recovery	Latch-off; requires input power cycle	—
Indicators	Green = output OK, red = fault, red = over-temp	see Table 9

Table 6: Electrical characteristics (design targets).

3.5 Connector pinout

Designator	Signal	Note
IN1	Input 1 positive	M3 solder-on threaded standoff
IN1	Input 1 return	M3 solder-on threaded standoff
IN2	Input 2 positive	M3 solder-on threaded standoff
IN2	Input 2 return	M3 solder-on threaded standoff
OUT	Output positive (V+)	M3 solder-on threaded standoff
OUT	Output return (GND)	M3 solder-on threaded standoff

Table 7: I/O connections. **[TODO: confirm silkscreen labels once the layout is done.]**

3.6 Mechanical & environmental

Parameter	Value / requirement
PCB outline	45 mm × 75 mm max
Mounting holes	4 × 3 mm dia., 70 mm × 40 mm pattern
Bottom thermal pad	45 mm × 40 mm, contact with enclosure floor
Assembled height	< 13.5 mm target; 18 mm fallback ceiling
Indicators	Bottom side, output end

Table 8: Mechanical requirements.

3.7 Indicator logic

Green (power)	Red (fault)	Red (over-temp)
ON	OFF	OFF
OFF	ON	OFF
OFF	OFF	ON

Table 9: Indicator truth table. **[TODO: add state column once LED wiring is final.]**

3.8 Typical application

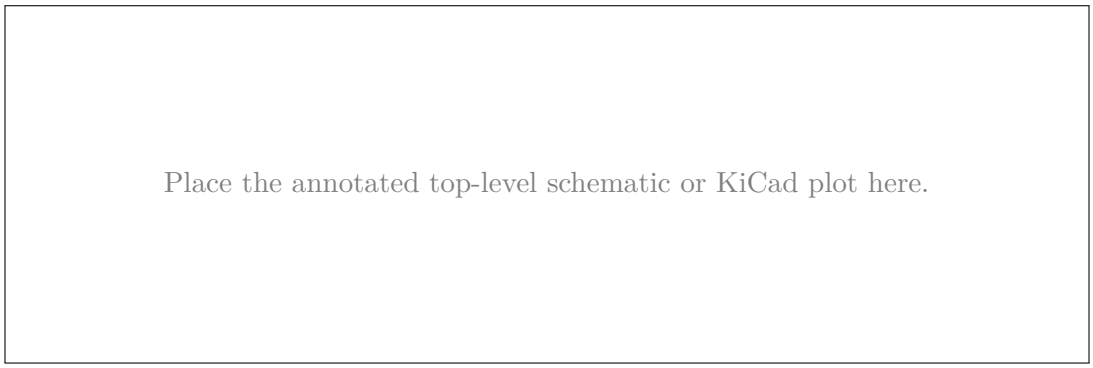


Figure 2: Typical application circuit. **[TODO: replace placeholder.]**

4 Design Calculations

Note — All equations are from the **LT8390 datasheet, Rev. F** (Applications Information, pp. 17–25) unless stated otherwise. Values tagged **(provisional)** are carried over from `BuckBoostTopology.md` and must be re-verified once the schematic is frozen.

4.1 Input stage — priority mux and protection

- Priority controller: LTC4417 (or equivalent) — fixed pin-assigned priority, Input 2 wired to the highest-priority pin.
- Per-channel UVLO: program the controller's undervoltage threshold to 11 V. **[TODO: insert resistor calculation from the LTC4417 datasheet: V_{UV} threshold and hysteresis formulas, show R_{top}, R_{bot}.]**
- Fuse sizing (each input line, ahead of the mux):

$$I_{FUSE} \geq \frac{P_{OUT}}{\eta \cdot V_{IN,min}} k_{margin} \tag{1}$$

with the worst-case input current evaluated at $V_{IN,min} = 11\,\text{V}$ and the burst power ($24\,\text{V} \times 8\,\text{A} = 192\,\text{W}$). Margin factor k_{margin} = **[TODO: 1.5–2, to be confirmed]**. **[TODO: finalize after the efficiency estimate (subsection 4.9)]**.

4.2 Buck-boost power stage

4.2.1 Switching frequency

$R_{12} = 226\,\text{k}\Omega$ (1 %) selects $f_{SW} = 200\,\text{kHz}$ (datasheet Table 1). Spread spectrum ON (±15 % modulation) for lower EMI. Rationale: thermal/size compromise; see Appendix C.

4.2.2 Inductor ripple current

$$\Delta I_{L(BUCK)} = \frac{V_{OUT}\,(V_{IN(MAX)} - V_{OUT})}{f_{SW}\,L\,V_{IN(MAX)}} \tag{2}$$

$$\Delta I_{L(BOOST)} = \frac{V_{IN(MIN)}\,(V_{OUT} - V_{IN(MIN)})}{f_{SW}\,L\,V_{OUT}} \tag{3}$$

Worked worst case (11 V in, 24 V out, $L = 10\,\mu\text{H}$, $f_{SW} = 200\,\text{kHz}$):

$$\Delta I_{L(BOOST)} = \frac{11\,\text{V}\,(24\,\text{V} - 11\,\text{V})}{200\,\text{kHz} \cdot 10\,\mu\text{H} \cdot 24\,\text{V}} \approx 2.98\,\text{A} \tag{4}$$

4.2.3 Inductor selection

$$L_{BUCK} > \frac{V_{OUT}\,(V_{IN(MAX)} - V_{OUT})}{f_{SW}\,I_{OUT}\,\Delta I_L\%\,V_{IN(MAX)}} \tag{5}$$

$$L_{BOOST} > \frac{V_{IN(MIN)}^2\,(V_{OUT} - V_{IN(MIN)})}{f_{SW}\,I_{OUT}\,\Delta I_L\%\,V_{OUT}^2} \tag{6}$$

$$\text{slope-comp. stability:} \quad L > \frac{10\,V_{OUT}\,R_{SENSE}}{f_{SW}} \approx 2.4\,\mu\text{H} \tag{7}$$

Chosen: $L_1 = 10\,\mu\text{H}$, ≥18 A saturation, ferrite, low DCR. Height-budget tradeoff: the 11 V→24 V, 8 A corner peaks at ≈19 A — accepted, see risk log (Appendix D).

4.2.4 Inductor current sense and peak limit

$$R_{SENSE(BUCK)} = \frac{2 \cdot 50\,\text{mV}}{2\,I_{OUT} + \Delta I_{L(BUCK)}} \tag{8}$$

$$R_{SENSE(BOOST)} = \frac{2 \cdot 50\,\text{mV} \cdot V_{IN(MIN)}}{2\,I_{OUT}\,V_{OUT} + \Delta I_{L(BOOST)}\,V_{IN(MIN)}} \tag{9}$$

Worst corner (11 V in, 24 V out, 8 A): $R_{SENSE(BOOST)} \approx 2.64\,\text{m}\Omega$. Chosen $R_5 = 2\,\text{m}\Omega \Rightarrow$ **24 % margin** (datasheet requires final value 20–30 % below calculated). Peak limit: $I_{PEAK} = 50\,\text{mV}/R_5 = 25\,\text{A}$.

4.2.5 Output CC limit and monitor

With CTRL tied to VREF (≥ 1.35 V), the ISP/ISN threshold is the full 100 mV scale:

$$I_{LIM} = \frac{100\,\text{mV}}{R_{11}} = \frac{100\,\text{mV}}{12\,\text{m}\Omega} = 8.33\,\text{A} \tag{10}$$

Monitor output: $V_{ISMON} = 10\,V_{(ISP-ISN)} + 0.25\,\text{V}$. **[TODO: transcription re-checked against the LT8390 datasheet p. 22.]**

4.2.6 Capacitors

- **Input:** RMS ripple current in the buck region $I_{IN,RMS} \approx I_{OUT}\sqrt{D(1-D)}$, $D = V_{OUT}/V_{IN}$; worst case $D = 0.5$ gives $I_{OUT}/2 \approx 4\,\text{A}$ at the 8 A burst rating — shared by `C11`–`C14` (4 × 10 µF, 50 V X7R/X7S, 1210). Plus `C8` = 1 µF at the VIN pin.
- **Output:** `C19`–`C22` (4 × 10 µF, 50 V) + `C23` (180 µF polymer); `C24` = 1 µF at the VOUT pin.
- **Bootstrap:** `C10`, `C18` = 100 nF X7R (≥100× top-FET gate charge; datasheet range 0.1 µF to 0.47 µF).
- **INTVCC:** `C15` = 4.7 µF plus a second 1 µF right at the pin (datasheet layout recommendation).

4.2.7 MOSFET selection

`Q3`–`Q6` = NCEP40T11G (logic-level; gate drive is the 5 V INTVCC rail). $R_{DS(ON)} \approx 3.3\,\text{m}\Omega$ typ. at $V_{GS} = 4.5$ V. 40 V BVDSS on a 36 V rail leaves ≈11 % ringing headroom — bench-verify, snubber if needed.

4.3 Feedback divider and output voltage select

$$V_{OUT} = 1.00\,\text{V} \times \frac{R_{top} + R_{bot}}{R_{bot}}, \qquad R_{bot} = R_{17} = 10.0\,\text{k}\Omega \tag{11}$$

$$12\,\text{V}: \quad R_{top} = R_{15} = 110\,\text{k}\Omega \tag{12}$$

$$24\,\text{V}: \quad R_{top} = R_{15} + R_{16} = 230\,\text{k}\Omega \tag{13}$$

`J1` shunts `R16`: **fitted = 12 V, removed = 24 V**; a missing jumper defaults to 12 V (fails low, no runaway).

4.4 Enable / UVLO

$$V_{IN(turn\text{-}on)} = 1.233\,\text{V} \cdot \frac{R_8 + R_9}{R_9} + 2.5\,\mu\text{A} \cdot R_8$$
$$V_{IN(turn\text{-}off)} = 1.220\,\text{V} \cdot \frac{R_8 + R_9}{R_9} \tag{14}$$

With $R_8 = 365\,\text{k}\Omega$, $R_9 = 56.2\,\text{k}\Omega$: ≈10.2 V **on,** ≈9.1 V **off** — deliberately below the LTC4417's 11 V floor so the mux arbitrates first.

4.5 Soft-start and compensation

$$t_{SS} = \frac{1\,\text{V} \cdot C_9}{12.5\,\mu\text{A}} = \frac{1\,\text{V} \cdot 100\,\text{nF}}{12.5\,\mu\text{A}} \approx 8\,\text{ms} \tag{15}$$

`R10` = 499 kΩ (SS→VREF) selects **latch-off** on fault. Compensation `R13` = 15 kΩ + `C25` = 15 nF; bench-tune with load steps at both outputs. Constraint: $C_9 \geq 5$–$10 \times C_{25}$ (ratio here: 6.7×, OK).

4.6 Over-temperature latch

Trip point set by the NTC divider against the TL431's 2.495 V reference, fed from a 5.1 V zener rail so the trip point barely moves with battery voltage:

$$R_{NTC}(trip) = R_{21}\left(\frac{V_Z}{2.495} - 1\right) = 1.33\text{k}\left(\frac{5.1}{2.495} - 1\right) \approx 1.39\,\text{k}\Omega \tag{16}$$

⇒ trip ≈85 °C nominal, ±8 °C realistic (zener tolerance, TL431 grade, NTC B-tolerance). `TH1` (10 k NTC) must be mounted on the hot copper, not in free air. Latch resets only when input power is cycled.

4.7 Indicator resistors

Continuous worst case is 36 V input; sizes chosen so no resistor exceeds ≈50 % of its package rating:

Ref	Value	Dissipation (worst case)	Package
`R18` (green)	4.7 kΩ	≈0.10 W	1206
`R19` (red)	6.8 kΩ	≈0.17 W	1206 min., 2010 preferred
`R29` (red)	6.8 kΩ	≈0.17 W	1206 min., 2010 preferred

Table 10: LED series resistors. **[TODO: re-check currents against final LED V_F bins.]**

4.8 Thermal analysis

- **INTVCC self-heating** (≈1.6 W inside the IC at 36 V, 200 kHz, four ≈50 nC gates):

$$P = V_{IN} \cdot I_{INTVCC} \approx 36 \cdot 45\,\text{mA} \approx 1.6\,\text{W}, \qquad T_J = T_A + P\,\theta_{JA} \tag{17}$$

  [TODO: verify θ_{JA} for the chosen layout and confirm T_J at max ambient; fallback options: 150 kHz, lower-Q_g FETs.]
- **Bottom thermal pad:** 45 mm×40 mm exposed pad, vias from `U1` exposed pad and the power MOSFETs; contact with the enclosure floor. **[TODO: add measured case temperature once available.]**
- **Heatsinks:** low-profile finned heatsinks on `U1` and the switching MOSFETs where height allows.

4.9 Efficiency budget

[TODO: fill from bench measurement or a loss budget (conduction + switching + gate charge). One row per corner:]

Corner	Input	Output
Buck, 36 V → 12 V, 2 A	**[TODO: %]**	**[TODO: %]**
Buck-boost corner, 12 V in, 12 V out	**[TODO: %]**	**[TODO: %]**
Boost, 11 V → 24 V, 2 A	**[TODO: %]**	**[TODO: %]**

Table 11: Efficiency at nominal duty.

5 Verification & Test Plan

[TODO: expand into a full procedure once the first board is built.]

Test	Pass criterion
Input priority	Input 2 always selected when valid; clean fallback to Input 1
UVLO per channel	Both channels disable below 11 V, restart above it
Reverse polarity	No damage / no output with either input reversed
Output voltage	12 V and 24 V within ±1.5 % at nominal load
Output CC limit	Current limit at 8.33 A ± tolerance
Load transient	Stable at both outputs, no latch-off, acceptable deviation
Over-temp trip	Latch fires at target temperature; recovers only on power cycle
Thermal soak	T_J, case and enclosure-floor temperatures within limits at worst corner

Table 12: Verification test matrix.

6 Open Items & Risks

See the risk register in Appendix D.

A Bill of Materials

[TODO: Keep in sync with `SchematicBOM.md`; summarize the main parts here.]

Ref	Part / value	Purpose	Status
`U1`	LT8390 (TSSOP-28 EP)	Buck-boost controller	placed
`Q3`–`Q6`	NCEP40T11G	Power switches	placed
`L1`	10 µH, ≥18 A sat	Main inductor	placed
`R5`	2 m Ω	Inductor current sense	placed
`R11`	12 m Ω	Output current sense	placed
`U3` + `TH1`	TL431 + NTC	Over-temp comparator / sensor	**[TODO: verify]**
`Q7`–`Q11`, `D1`–`D4`, `J1`	**[TODO: parts]**	Latch, LEDs, output select	to add

Table 13: Principal components (excerpt).

B Equation Reference

Full derivations live in section 4; this is a quick-reference index.

Quantity	Equation	Source
Inductor ripple (buck)	$V_{OUT}(V_{IN(MAX)} - V_{OUT})/(fLV_{IN(MAX)})$	LT8390 p. 18
Inductor ripple (boost)	$V_{IN(MIN)}(V_{OUT} - V_{IN(MIN)})/(fLV_{OUT})$	LT8390 p. 18
Peak current limit	$I_{PEAK} = 50\,\text{mV}/R_{SENSE}$	LT8390 p. 18
Output CC limit	$I_{LIM} = 100\,\text{mV}/R_{11}$ (CTRL high)	LT8390 p. 22
Output voltage	$V_{OUT} = 1.00 \cdot (R_{top} + R_{bot})/R_{bot}$	LT8390 p. 21
UVLO	$V_{IN} = 1.233\,(R_8 + R_9)/R_9 + 2.5\,\mu\text{A}\,R_8$	LT8390 p. 21
Soft-start	$t_{SS} = 1\,\text{V} \cdot C_9/12.5\,\mu\text{A}$	LT8390 p. 21
NTC trip resistance	$R_{NTC} = R_{21}(V_Z/2.495 - 1)$	TL431 datasheet
NCV/**[TODO:]**	**[TODO:]**	**[TODO:]**

Table 14: Equation quick reference.

C Design Decisions Log

Decision	Date
LT8390 4-switch buck-boost: single inductor, input straddles both outputs, PGOOD + latch-off available.	2026-09
200 kHz switching: thermal/size compromise; spread spectrum on.	2026-09
10 µH/18 A-sat inductor: height budget; accepted 19 A corner peak.	2026-09
Jumper presets instead of trim pot: no drift, no knob to bump, lower height.	2026-09
Latch pulls EN/UVLO: assured full shutdown; resets only on power cycle.	2026-09
Two red LEDs (fault vs. over-temp): distinguishable states, simpler than steered diode.	2026-09
[TODO: append each new decision with rationale and date.]	

Table 15: Design decision record.

D Open Items & Risk Register

Item	Severity	Status
Analog support blocks (`Q7`–`Q11`, `D1`–`D4`, `J1`, `TH1`) not yet placed	high	open
No footprints assigned on any symbol	high	open
Fuse ratings not finalized	medium	open
INTVCC self-heating (≈1.6 W) — T_J margin unverified	high	open
19 A corner peak vs. 18 A inductor saturation	medium	accepted
40 V FETs on 36 V rail — ringing headroom	medium	bench-check
Height budget: PCB + parts + heatsink under 13.5 mm	medium	open

Table 16: Open items carried from the current schematic audit.

E Revision History

Rev	Date	Author	Change
0.1	October 5, 2026	**[TODO: name]**	Initial template created
[TODO:]	**[TODO:]**	**[TODO:]**	**[TODO:]**

Table 17: Revision history.

F References

[1] LT8390 datasheet, Rev. F — `Datasheets/LTC8390.pdf`
[2] LTC4417 datasheet — `Datasheets/LTC4417.pdf`
[3] NCEP40T11G datasheet — `Datasheets/NCEP40T11G.pdf`
[4] TL431 datasheet — **[TODO: download to `Datasheets/agentDownload/`]**
[5] Project specification — `Specifications/SpecSheet.pdf`
[6] **[TODO: add any further references as they are used.]**